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MSDS 456

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3/12/2026

NFL Late-Game Decision Support Tool Technical Documentation

1. Dataset Description

This project uses NFL play-by-play data gathered from the nflreadr R package, which provides a regularly updated version of the NFL's public play-by-play dataset. The dataset contains detailed records for every play from each NFL game.

For this project, play-by-play data from the 2015 to 2025 seasons were downloaded using:

nflreadr::load_pbp(seasons)

Each observation in the dataset represents a single play within a game. The original play-by-play dataset contains hundreds of variables describing game context, team information, play outcomes, and advanced metrics.

For the purposes of this analysis, a subset of variables was retained to focus specifically on late-game decision-making scenarios. The final analysis dataset (simulator_base) contains 51,477 observations and 31 variables after filtering and data manipulation.

Key Variables used:

Variable	Description
season	NFL season year
game_id	Unique identifier for each game
play_id	Sequential play identifier within a game
drive	Drive number within the game
posteam	Offensive team on the play
defteam	Defensive team on the play
down	Current down
ydstogo	Yards needed for a first down
yardline_100	Distance from the opponent's goal line
score_differential	Offensive team score minus defensive team score
game_seconds_remaining	Seconds remaining in regulation
half_seconds_remaining	Seconds remaining in the half
play_type	Raw play classification
epa	Expected Points Added for the play
desc	Text description of the play

Additional derived variables were created during preprocessing to support scenario analysis.

2. Data Processing & Assumptions

The data process used was designed to transform the raw play-by-play dataset into a structure suitable for analyzing late-game offensive decision making.

Late-Game Situation Filtering

The analysis focuses exclusively on situations occurring in the final two minutes of a half or game. Two situation categories were defined:

- End of Half:
  - Plays occurring in the second quarter with 120 seconds or fewer remaining in the half.
- End of Game:
  - Plays occurring in the fourth quarter with 120 seconds or fewer remaining in regulation.

Plays outside these conditions were excluded from the analysis.

Play Type Mapping

Raw play types from the play-by-play data were mapped into a simplified set of five actionable play categories relevant to late-game decision making:

Play Bucket	Included Raw Play Types
Run	rushing plays
Pass	passing plays
Spike	quarterback spikes
Kneel	quarterback kneels
Field Goal Attempt	field goal attempts

Other play types (penalties, punts, etc.) were excluded because they are not primary offensive decisions in typical two-minute scenarios.

Time Elapsed Calculation

To estimate the amount of clock time consumed by each play, the difference between the current play's remaining game time and the next play's remaining time was calculated. This produced the variable:

sec_elapsed_next_play

This value represents the seconds burned between consecutive plays.

To prevent erroneous values caused by stoppages or quarter transitions, only values between 0 and 60 seconds were retained.

Drive Outcome Identification

Drive-level outcomes were derived by grouping plays by game_id and drive. Each drive was classified into one of four outcomes:

Outcome	Definition
TD	Drive resulted in a touchdown
FG	Drive resulted in a made field goal
Turnover	Drive ended with interception or lost fumble
No Score	Drive ended without scoring

Binary indicators were also created for:

- touchdown probability
- field goal probability
- turnover probability
- probability of any score

These metrics allow the simulator to estimate the likelihood of various outcomes following a particular play choice.

Situation Bucketing

To improve interpretability and maintain sufficient sample sizes, several continuous variables were grouped into categorical buckets.

Time Remaining

Bucket	Seconds Remaining
0-30	≤ 30 seconds
31-60	31-60 seconds
61-90	61-90 seconds
91-120	91-120 seconds

Score Differential

Bucket	Description
Down 8+	Trailing by 8 or more
Down 4-7	Trailing by 4-7
Down 1-3	Trailing by 1-3
Tied	Score tied
Up 1-3	Leading by 1-3
Up 4-7	Leading by 4-7
Up 8+	Leading by 8 or more

Field Position

Bucket	Distance from Goal Line
1-20	Shortened Red zone
21-40	Near scoring/FG range
41-60	Midfield
61-80	Own territory
81-99	Deep own territory

These bucket definitions allow the simulator to compare historical outcomes under similar game contexts.

3. Modeling Techniques

This project does not implement a predictive machine learning model. Instead, it employs a historical decision-support framework based on observations.

For each combination of:

- situation type
- time bucket
- score bucket
- field position bucket
- next play choice

historical outcomes are summarized to estimate the expected outcome of each play type.

The approach can be described as a conditional probability model, where the probability of various drive outcomes is calculated using observed historical frequencies within each scenario bucket.

This approach was chosen because:

- It provides interpretable results suitable for coaching and decision analysis.
- It avoids overfitting issues that can arise in sparse late-game data.
- It allows users to directly inspect historical plays supporting each scenario.

4. Summarization Techniques

The main summarized dataset used by the simulator is simulator_summary_mvp.

This dataset aggregates plays by:

- situation_type
- time_bucket
- score_bucket
- yardline_bucket
- next_play_bucket

For each group, the following statistics are calculated:

Metric	Description
n	Number of historical plays
avg_sec_elapsed	Average seconds burned by the play
median_sec_elapsed	Median seconds burned
avg_yardline	Average field position
avg_epa	Average expected points added
p_td	Probability the drive results in a touchdown
p_fg	Probability the drive results in a field goal
p_any_score	Probability the drive results in any score
p_turnover	Probability the drive ends in a turnover

These summaries allow the Shiny application to present quick comparisons between play choices within the same scenario.

5. Technical Considerations

Several technical challenges arose during development.

Sparse Late-Game Data

Late-game scenarios are relatively rare, and some combinations of time, score, and field position had very few observations. To address this issue:

- variables were grouped into broader buckets
- sample size warnings were included in the app when observations were limited

This ensured that simulator outputs remained statistically meaningful.

Interactive Drive Exploration

To support deeper analysis, the application allows users to select a historical play and view the remainder of that drive from that point forward. This required linking the displayed play to the original game_id, drive, and play_id fields and dynamically retrieving the corresponding sequence of plays.

This feature enables users to contextualize simulator recommendations within real game situations.